



Machine Learning–Enabled Microalgae Bioprocess Optimization for Alternative Foods and Water Sustainability

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Abstract

Purpose of Review Global food insecurity and increasing freshwater scarcity continue to intensify as a result of population growth, urbanisation, climate variability, and the depletion of natural resources. Addressing these interconnected challenges requires a transition away from conventional, resource-intensive food systems toward sustainable protein alternatives that can deliver adequate nutrition with a reduced freshwater demand. Microalgae have emerged as a strong candidate in this context due to their rapid growth rates, broad environmental tolerance, ability to utilise carbon dioxide, and capacity to grow in brackish water, seawater, or nutrient-rich wastewater, thereby substantially lowering their blue-water footprint. While several microalgal species are already produced as single-cell protein (SCP), large-scale deployment across food, health, and industrial applications remains limited by economic, technical, and operational constraints. This review critically evaluates the potential and limitations of microalgae as a scalable solution for food and water security. Considering the increasing digitalisation of biomanufacturing, particular attention is given to the role of computational biology and artificial intelligence (AI)–enabled strategies in overcoming cultivation and process optimisation bottlenecks.

Recent Findings Recent advances demonstrate that artificial intelligence (AI) approaches, particularly machine learning (ML), alongside Internet of Things (IoT)–based sensing, can significantly improve resource-use efficiency and nutrient recovery in microalgae production systems. In parallel, the growing application of multi-omics and systems biology tools is generating high-resolution datasets that are increasingly important for the development, validation, and deployment of robust ML models.

Summary This review distinguishes itself from previous studies by presenting an integrated perspective that links alternative protein production with environmental sustainability, particularly within the Water–Food–Energy nexus, while systematically examining ML applications across the microalgal bioprocess value chain. Key knowledge gaps, future research priorities, and the practical challenges associated with implementing AI-driven solutions in microalgae-based systems are also critically discussed.

Highlights

- Microalgae are a sustainable alternative protein addressing food security and freshwater scarcity.
- Artificial intelligence, particularly machine learning-based microalgal cultivation optimization, resource efficiency, and nutrient recovery.
- Digital twins, IoT monitoring, and predictive control reduce economic and operational barriers to scale-up.
- AI-enabled bioprocessing links alternative proteins to sustainability within the Water–Food–Energy nexus.

Keywords Microalgae · Machine learning · Artificial intelligence · Alternative foods · Nutrient recovery · Omics · IoT · Digital twins

Introduction

The global demand for protein-rich foods and freshwater has increased steadily over recent decades, largely due to population growth, rapid urbanisation, and the continued depletion of natural resources. By 2050, the global population is expected to approach 10 billion, with food demand, particularly for meat, projected to increase by approximately 73% [1]. Across most regions, protein intake has traditionally relied on animal-derived products such as meat, eggs, and dairy. However, contemporary food systems are increasingly exposed to external pressures, including climate change, pandemics, and geopolitical instability, all of which can disrupt supply chains and intensify resource scarcity. These disruptions are likely to affect agricultural productivity across successive growing seasons, placing additional strain on already stressed food systems. As a result, current agricultural practices and food distribution models are unlikely to meet future nutritional requirements without substantial transformation. Achieving the United Nations Sustainable Development Goal on food security (SDG 2) will therefore require a transition toward alternative protein sources that are nutritionally adequate, scalable, and environmentally sustainable [2].

In response to these challenges, proteins have gained increased attention as a central component of sustainable food production strategies. Plant-based protein sources, particularly cereals and oilseeds, currently supply up to 65% of global protein intake and represent the most widely adopted alternatives to animal-derived proteins [3]. Despite their advantages, these sources face several limitations, including suboptimal amino acid profiles, flavour constraints, high land and water requirements, and competition for arable land. The total global arable land available for crop production is projected to be approximately 1,661 million hectares by 2025 [4]. Moreover, cereals and oilseeds are extensively utilised in livestock and poultry feed, creating direct competition between food and feed systems and further challenging food security for human consumption [5]. These limitations highlight the need for alternative protein sources capable of meeting both human and animal nutritional demands without exacerbating environmental degradation. Single-cell organisms, including bacteria, yeast, and microalgae, have therefore emerged as promising alternatives to conventional plant- and animal-based proteins [6].

Within the broader discussion on sustainable protein systems, the term alternative foods refer to products derived from non-conventional sources such as microorganisms, insects, or plant-based analogues. These products are designed to supplement or replace animal-derived proteins while delivering measurable reductions in land use, freshwater consumption, and greenhouse gas emissions per kilogram of protein produced. In microalgae-based systems,

these products fall into three categories with distinct processing and regulatory implications. Whole-biomass products, such as dried *Spirulina* or *Chlorella* powders containing 46–68% protein, require only harvesting and drying but face sensory barriers including intense green colour and off-flavours that limit consumer acceptance. These products also demand stringent safety validation when cultivation uses wastewater [7]. Refined fractions and protein isolates, comprising concentrates with 50–70% protein and isolates exceeding 85% protein, require cell disruption and purification to preserve techno-functional properties such as foaming and emulsification. These properties make them suitable for meat analogues and dairy alternatives. Formulated food products combine microalgal components with other ingredients to create fortified pasta, snacks, or plant-based meats, where sensory masking strategies such as encapsulation and fermentation become critical for consumer acceptance. These distinctions determine whether AI and machine learning optimisation targets volumetric productivity, extraction efficiency, functional property retention, or sensory acceptance. They also dictate the safety validation required when cultivation integrates wastewater nutrient recovery.

Microalgae represent one of several alternative protein routes being explored to address the limitations of conventional animal-derived proteins. Other key routes include plant-based proteins from legumes and oilseeds, insect-derived proteins, and fermentation-based approaches including both biomass fermentation and precision fermentation. Plant-based proteins currently dominate the alternative protein market, while insect proteins offer efficient bio-conversion of organic side streams but face regulatory and consumer acceptance hurdles [8]. Precision fermentation uses engineered microorganisms to produce specific functional proteins such as dairy proteins, though the technology faces challenges in scaling costs and consumer acceptance of cell-modified ingredients. Importantly, these alternative routes are complementary rather than mutually exclusive. Microalgae cultivation can be integrated with fermentation processes, as demonstrated by recent work using bacterial fermentation to enhance protein extraction from *Chlorella pyrenoidosa*, achieving extraction yields of 35.45% while reducing energy consumption compared to conventional methods [9]. Additionally, formulated food products increasingly combine microalgae proteins with plant-based proteins to achieve improved nutritional profiles, functional properties, and sensory characteristics while masking undesirable flavours. This complementarity extends to sustainability outcomes: microalgae production on wastewater nutrients can supply protein-rich biomass for fermentation feedstocks or serve as a functional ingredient alongside precision-fermented proteins in hybrid food formulations, thereby supporting circular bioeconomy principles.

Alongside food security concerns, freshwater scarcity has become a parallel global challenge with significant implications for agricultural productivity and protein production pathways. Microalgae offer distinct advantages within the water–food–energy nexus, as they can be cultivated on non-arable land using brackish water, seawater, or nutrient-rich municipal and industrial effluents instead of freshwater [10]. Compared with conventional crops, microalgal cultivation can achieve a substantially lower blue-water footprint per unit of protein produced [11], while simultaneously enabling nutrient recovery through the removal of nitrogen, phosphorus, and organic pollutants from wastewater streams. In addition, microalgae exhibit high photosynthetic efficiency, broad ecological adaptability, and a comparatively low net carbon footprint. It has been estimated that microalgae contribute nearly 50% of global oxygen production [12], and that the production of one kilogram of algal biomass can capture approximately 1.83 kg of CO₂ [13]. Current estimates suggest the existence of between 200,000 and 800,000 species of photosynthetic microalgae, highlighting their extensive biodiversity and biotechnological potential [14].

Several microalgal species, including *Spirulina platensis*, *Spirulina maxima*, *Chlorella pyrenoidosa*, and *Chlorella vulgaris*, have been utilised as single-cell protein sources for several decades, with protein contents reaching up to 70% of dry biomass. In addition to proteins, microalgae produce lipids, carbohydrates, phycobiliproteins, and a wide range of secondary metabolites. Many of these compounds exhibit documented anticancer, antioxidant, antimicrobial, anticoagulant, antihypertensive, and immunomodulatory activities [15], enhancing the value of microalgae for food, feed, and health-related applications. Despite this potential, the large-scale deployment of microalgae-based bioprocesses remains limited by technical, economic, and operational challenges. Beyond their role as protein sources, microalgae–bacteria consortia used in wastewater treatment further demonstrate the multifunctionality of microalgal systems, enabling simultaneous nutrient removal and biomass generation. For instance, optimising microalgae–bacteria inoculation ratios during the start-up of bioreactors treating paper pulp wastewater, achieving high ammonium removal efficiencies alongside the production of lipid-rich photogranular biomass suitable for biofuel applications [16].

Commercial applications of microalgae as single-cell proteins (SCP) have historically been limited to a few species, including *Spirulina platensis*, *Chlorella vulgaris*, *Scenedesmus obliquus*, and *Dunaliella* sp., over several decades [5]. This limitation is largely attributed to challenges encountered during bioprocessing, which encompasses both upstream and downstream phases [17, 18]. Upstream processing is constrained by strain-specific characteristics, growth conditions,

and bioreactor design complexities. Given that the costs of photobioreactors (PBRs), growth media, and biomass yield strongly influence overall production costs, reducing the cost of microalgal biomass remains a significant challenge. Downstream processing, including harvesting, drying, and purification of valuable bioproducts, is energy-intensive and further adds to production costs. To address these challenges, digitized approaches in microbial bioprocessing are increasingly being adopted [19]. Integration of artificial intelligence (AI) technologies, including machine learning (ML), alongside Internet of Things (IoT) systems into microalgae bioprocesses enables sophisticated data analytics, facilitating the identification of process optimizations and cost-effective solutions [20].

Several recent reviews have explored microalgae for food, biofuel, or biorefinery applications, while others have focused on ML tools for cultivation optimisation. However, these studies typically address these themes in isolation and rarely contextualise digital technologies within broader sustainability frameworks. This review integrates evidence across the entire microalgal bioprocessing chain, with a particular focus on their potential as alternative protein sources and their incorporation into sustainable food and water systems. A central and novel objective is to critically examine the convergence of microalgae-based food production with wastewater treatment as a circular economy strategy, where nutrient-rich effluents serve as cultivation media for microalgae that are subsequently processed into safe food ingredients. This integrated perspective explicitly addresses the technical feasibility, safety requirements, regulatory pathways, and economic considerations for such systems. Unlike prior studies, we link upstream and downstream bioprocess stages and examine ML applications across multiple functions, including process prediction, control, strain selection, digital twins, and quality assessment. These are further framed within the water–food–energy nexus, emphasising wastewater treatment, nutrient recovery, and resource efficiency. Additionally, the review synthesises comparative tables of ML models, identifies key data and methodological gaps, and introduces a conceptual hybrid AI–physics framework to guide future process optimisation. By integrating these dimensions, this work offers a comprehensive, problem-oriented perspective that is not provided by existing reviews.

Methodology

This review was conducted using a structured literature search to identify studies on artificial intelligence and machine learning applications in microalgae systems related to alternative food production and wastewater-based nutrient

recovery. The Scopus database was used as the primary source due to its broad coverage of peer-reviewed scientific literature. The search was performed using combinations of keywords including “microalgae”, “machine learning”, “wastewater treatment”, “microalgae protein”, and “bioprocess optimisation”. Publications from 2000 to 2026 were considered. The initial search results were screened based on title, abstract, and relevance to AI-driven microalgae cultivation, monitoring, strain optimisation, or downstream processing. Studies that did not involve microalgae systems or did not address AI or machine learning applications were excluded. The selected studies were then analysed and categorised according to their primary application areas, including growth prediction, process optimisation, monitoring and control, strain development, and downstream processing.

Overview of Microalgae Bioprocess and Challenges

Microalgae bioprocesses comprise upstream and downstream processing stages. Upstream processing involves strain selection and cultivation under controlled conditions (temperature, pH, light intensity, CO₂ supply, nutrient availability) that vary widely between species and growth modes (autotrophic, mixotrophic, heterotrophic) [21]. Variations in cultivation conditions result in species-specific and non-generic cultivation methods [20]. Every species of microalgae grows best under optimum conditions, including temperature, pH, CO₂ concentration, aeration, light intensity, and photo-dark time. These conditions are optimised to fit the needs of every microalgae species [22, 23] and are necessary for their growth. Cultivating microalgae-bacteria consortia in industrial effluents further confirms how upstream decisions such as inoculum composition and light regime simultaneously influence granule structure, chlorophyll and carotenoid content, lipid accumulation, and nitrogen removal efficiency, complicating manual optimisation of these systems [16].

Microalgae require essential nutrients for optimal growth, which directly influences both biomass yield and production costs. Open raceway ponds and closed photobioreactors (PBRs) are the two main cultivation systems used for large-scale microalgae production. While open raceway ponds are cost-effective, they are prone to contamination and evaporation losses, often resulting in lower and less predictable biomass yields [24]. Nevertheless, raceway ponds remain the dominant commercial cultivation system for food-grade microalgae production due to their relatively low capital investment and operational requirements, and they are widely used for large-scale cultivation of species such as *Arthrospira* (*Spirulina*) and *Chlorella* for dietary

supplements and functional food ingredients. Closed PBRs, in contrast, provide greater control over cultivation conditions but involve substantially higher capital and operating costs, which limits their commercial application primarily to high-value products such as astaxanthin from *Haematococcus pluvialis* rather than bulk protein production [25]. Relatively few studies have addressed the economic feasibility of large-scale microalgae cultivation using either open or closed PBR systems [26, 27]. Recent research on microalgae-bacteria consortia across different reactor configurations has demonstrated how operational parameters and hydrodynamics influence both biomass production and nutrient removal in wastewater treatment, highlighting the critical link between bioprocess design and water sustainability outcomes [28]. Specifically, variations in hydraulic retention time and slow-feeding strategies in algal-bacterial aerobic granular sludge reactors have been shown to significantly affect granule size, microbial community composition, and phosphorus and nitrogen removal efficiency [29].

A crucial phase in the synthesis of useful bioproducts for food, feed, and health uses is the downstream processing of microalgae biomass. Downstream processing includes harvesting, drying, cell disruption, and product purification [30]. Following microalgae cultivation, harvesting, and purification processes directly affect the quality and productivity of the produced biomass [31]. Microalgal biomass is harvested by means of several techniques, including evaporation, centrifugation, microfiltration, gravity sedimentation, and flotation. These techniques present trade-offs in efficiency, energy demand, technological feasibility/scalability, and product quality [32]. The targeted product and the features of the microalgae size, species, and cell density help to choose the best harvesting technique. In addition to lowering transportation costs, the drying method maintains the chemical integrity of microalgal biomass by preventing microbial contamination. Standard drying techniques consist of spray drying, freeze drying, solar drying, and drum drying [33]. The desired use will determine the drying method used. The purification stage uses mechanical, chemical, and biological approaches to disrupt microalgal cell walls in order to extract important compounds, e.g., proteins and bioactives [34]. Recent developments underline pretreatment methods such as biosurfactants or dispersers that improve cell wall disruption while preserving product integrity [35]. The downstream process struggles with cost efficiency and contaminated product; no one approach has been shown to be practical at the industrial level [36].

These upstream and downstream challenges are further amplified when microalgal biomass is intended for use as an alternative food or feed rather than for biofuel production, particularly in wastewater-integrated systems that are otherwise highly promising for resource

efficiency and water sustainability. For applications in human food or animal feed, significantly stricter standards must be met compared to biofuel or cosmetic uses. Only food-grade or GRAS-listed coagulants, flocculants, solvents, materials, and cleaning procedures are permissible. Regulatory frameworks, such as EU Novel Food authorization (Regulation 2015/2283), FDA GRAS status, and EFSA evaluations, mandate demonstrated absence of phycotoxins, minimal heavy-metal concentrations, and strict microbiological limits (typically total plate count $< 10^5$ CFU g⁻¹ and absence of pathogens such as *Salmonella* and *Listeria*). Moreover, persistent off-flavours (e.g., geosmin or 2-methylisoborneol) and the intense green color from residual chlorophyll continue to pose substantial barriers to consumer acceptance in meat or dairy alternatives and functional foods [5]. Any decolorizations or deodorization steps must preserve the nutritional profile and avoid the use of non-GRAS substances.

These food and feed-specific constraints severely restrict the choice of harvesting, drying, and extraction methods. Many inexpensive chemical flocculants that are routinely employed in biofuel production are not permitted, forcing reliance on gentler alternatives such as bio-flocculation, pH-induced flocculation, or membrane filtration [37]. When cultivation is coupled with municipal or industrial wastewater, extra rigorous validation becomes essential to ensure that no contaminants (heavy metals, pathogens, or micropollutants) are transferred to the final food-chain product. Artificial intelligence (AI), particularly its machine learning (ML) subset, alongside IoT-enabled monitoring are increasingly vital for overcoming these challenges while preserving food-grade compliance and water-treatment performance [38]. Predictive models can optimise growth conditions to maximise protein content and minimise toxin/off-flavour precursors; real-time sensors can trigger harvesting at the ideal window; and hybrid AI-physics frameworks can forecast granule dynamics in wastewater-fed consortia, reducing or eliminating chemical inputs. By intelligently integrating digital tools with wastewater-based cultivation, microalgae bioprocesses can deliver safe, high-quality alternative proteins alongside substantial nutrient recovery and water reuse benefits.

Artificial Intelligence and Machine Learning in Bioprocess Optimization

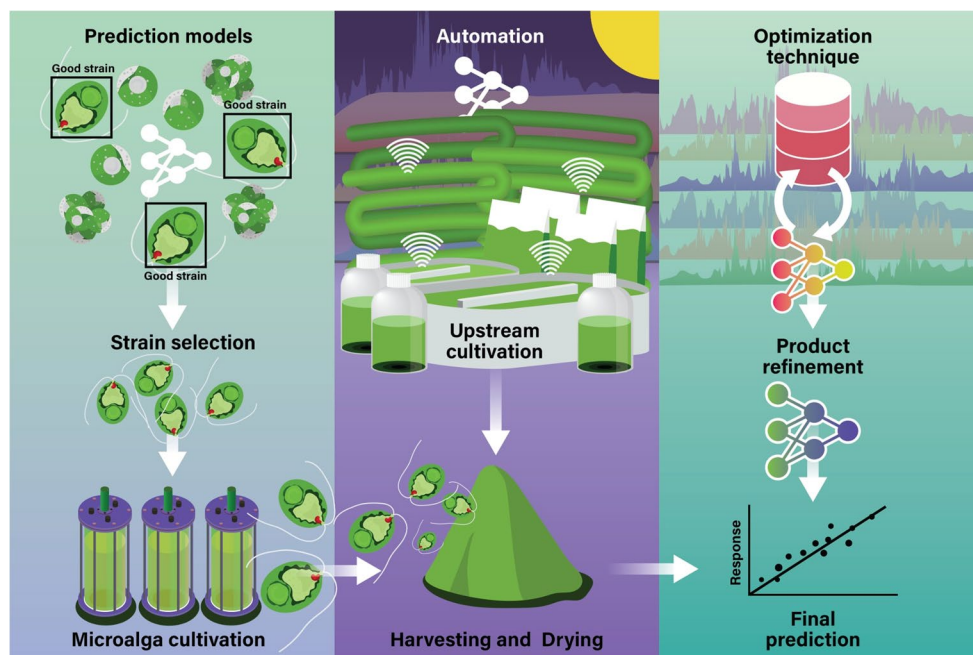
Artificial intelligence (AI) and machine learning (ML) are fundamentally reshaping microalgae bioprocessing by providing powerful tools to tackle longstanding industrial challenges. The integration of these computational approaches

into cultivation and downstream operations has opened new avenues for researchers and industry practitioners alike, enabling marked gains in efficiency and productivity through optimised growth conditions, enhanced real-time monitoring, and adaptive process control. AI-driven predictive analytics, in particular, allow for the early identification of operational bottlenecks and environmental stressors, thereby supporting more stable and reliable microalgae cultivation [39]. Extending these optimisation frameworks to microalgae-bacteria granular systems, such as the algal-bacterial aerobic granular sludge treating low-nitrogen pulp mill wastewater, would allow ML models to exploit the documented relationships between hydraulic regimes, granule morphology, and nutrient removal to identify robust operating windows and anticipate process instability [29].

AI technologies, particularly machine learning, enable creative developments with the optimization of products obtained from microorganisms. Using these technologies, researchers choose and modify microalgae strains with desired traits, including increased lipid content and pigment yields and higher biomass production. Combining AI-driven technology has enhanced downstream operations as well as product development for commodities produced from microalgae that generate higher product yield and quality of biofuel, nutraceuticals, and cosmetic materials. Microalgae bioprocessing has reached a turning point by way of artificial intelligence and machine learning integration, as this combination permits data-based decisions with live optimization and the identification of heretofore unexplored possibilities. These innovative technologies enable industrial players and researchers to establish new prospects in microalgae-based applications, thereby supporting sustainable, effective bioprocessing solutions [40]. The overall integration of AI methods, particularly machine learning, across the microalgae bioprocess pipeline from strain selection to downstream processing is illustrated in Fig. 1, highlighting key data inputs and decision points across upstream and downstream operations.

To make the diverse ML applications easier to follow, they can be grouped into four main tasks. In growth modelling and product prediction, artificial neural networks (ANN), random forests, and support vector machines are widely used to forecast biomass concentration, protein, lipid, or pigment content from cultivation parameters, with reported accuracies typically ranging from 90 to 98% [38, 41] on laboratory data (Table 1). Table 1 summarises representative ML applications used in microalgae bioprocess optimisation. The listed studies are categorised according to the four functional task groups. The table also indicates approximate technology readiness levels (TRLs) to provide an overview of the technological maturity of the reported approaches. As shown in Table 1, artificial

Fig. 1 Flow chart of ML integration throughout the microalgae bioprocessing pipeline



neural networks dominate the listed applications (over half the studies), reflecting their effectiveness on non-linear biological data, while genetic algorithms are mainly for optimisation tasks. However, gaps include underuse of interpretable models and limited studies on food-specific quality parameters like protein profiles or heavy-metal exclusion, with most relying on small lab datasets that hinder transfer to wastewater systems. While prediction accuracy is frequently reported in ML studies as a primary performance metric, it should not be considered in isolation when evaluating practical bioprocess applications. In industrial microalgae systems, the effectiveness of ML models must ultimately be assessed through process-level performance indicators such as biomass productivity, energy consumption, nutrient removal efficiency, and operational stability. Consequently, many studies listed in Table 1 report high predictive accuracy under laboratory conditions but provide limited evaluation of how these models translate into improvements in large-scale process performance. This limitation highlights an important research gap between laboratory-scale ML model development and validation under realistic industrial operating conditions. Process optimisation employs genetic algorithms and Bayesian optimisation, often combined with ML surrogate models, to identify the best nutrient levels, CO₂ concentration, light regimes, or harvesting timing that maximise yield or minimise energy use [47]. Real-time monitoring and control rely on IoT sensors that feed live data (pH, dissolved oxygen, turbidity) into soft sensors and model-predictive controllers, which automatically adjust aeration, lighting, or harvesting triggers while

detecting faults early [55]. Finally, product quality and downstream prediction use ML models to forecast final protein content, pigment levels, heavy-metal uptake, or off-flavour risk from upstream conditions, enabling proactive adjustments before harvesting.

Purely data-driven black-box models often fail when extrapolated to new conditions, such as different reactor designs, variable wastewater compositions, seasonal fluctuations, or novel strains. Hybrid or physics-informed models address this limitation by embedding established physical laws such as light-transfer equations, mass and energy balances, or Droop nutrient-quota kinetics directly into the machine learning architecture, or by using ML to correct the residuals of a mechanistic model. This approach offers four key advantages: (i) substantially reduced data requirements, (ii) improved generalization and extrapolation capabilities, (iii) physically consistent predictions (e.g., biomass concentrations cannot become negative), and (iv) enhanced regulatory acceptability, which is particularly critical for food-grade protein production. For instance [56], combined classical kinetic growth equations with ML residual correction to achieve robust performance in semi-continuous outdoor cultivation. Similar hybrid strategies are now being applied to predict granule dynamics, settling behavior, and nutrient removal efficiency in wastewater-fed microalgae-bacteria consortia, directly relevant to the systems described by Sátiro et al. (2024) [16].

Recent advances in artificial intelligence (AI), particularly machine learning (ML), have significantly enhanced the management and monitoring of microalgae cultivation. AI-powered systems analyse real-time data from key metrics

Table 1 ML models used in microalgae bioprocess optimisation categorized by functional task group and technological maturity

Model type	Application area	Input data	Output/Prediction	Accuracy	TRL	References
Growth/Product prediction						
Artificial Neural Network (ANN)	Prediction of lipid content in <i>Chlorella minutissima</i>	Wastewater concentration	Lipid percentage	96.34%	2–3	[38]
Convolutional Neural Network (CNN)	Carbohydrate accumulation in the cyanobacterial community	Carbon source	Carbohydrate content	89.66%	2–3	[38]
Artificial Neural Network (ANN)	Biomass prediction in <i>Chlorella vulgaris</i>	Nutrient levels	Biomass concentration	98%	2–3	[41]
Random Forest (RF)	Growth rate prediction in <i>Scenedesmus obliquus</i>	Growth factors	Growth rate	95%	2–3	[42]
Support Vector Machine (SVM)	Prediction of biomass in <i>Botryococcus braunii</i> cultivation	Growth conditions	Biomass concentration	92%	2–3	[43]
Artificial Neural Network (ANN)	Prediction of biomass in <i>Isochrysis galbana</i>	Growth conditions	Biomass concentrations	96%	2–3	[44]
Random Forest (RF)	Prediction of growth rate in <i>Phaeodactylum tricornutum</i>	Growth conditions	Growth rate	93%	2–3	[45]
Process optimization						
Support Vector Regression (SVR)	Biodiesel yield estimation from <i>Nannochloropsis oculata</i>	Oil to methanol ratio	Biodiesel yield	99.10%	2–3	[44]
Genetic Algorithm (GA)	Optimisation of nutrient concentrations for <i>Dunaliella salina</i>	Nutrient concentrations	Optimal growth conditions	Increase in growth rate by 15%	3–4	[46]
Genetic Algorithm (GA)	Optimisation of CO ₂ concentration for <i>Chlorella pyrenoidosa</i>	CO ₂ concentration	Growth conditions	Increase in growth by 20%	3–4	[47]
Artificial Neural Network (ANN)	Vacuum drying optimisation for <i>Chlorococcum infusionum</i>	Temperature, pressure	Moisture content	Lower error values from SVM	2–3	[48]
Support Vector Machine (SVM)	Vacuum drying optimisation for <i>Chlorococcum infusionum</i>	Temperature, pressure	Moisture content	Lower error values from ANN	2–3	[48]
Monitoring/Control						
Decision Tree (DT)	Classification based on morphology	Cell size, colour, shape	Species classification	92%	2–3	[45]
Decision Tree (DT)	Classification of microalgae species based on spectral data	Spectral reflectance values	Species classification	90%	2–3	[49]
Product quality/Downstream processing						
Artificial Neural Network (ANN)	Chlorophyll and carotenoid extraction from <i>Chlorella thermophila</i>	Different extraction parameters	Extraction yield	Not specified	2–3	[50]
Artificial Neural Network (ANN)	Prediction of astaxanthin content in <i>Haematococcus pluvialis</i>	Growth factors	Astaxanthin content	97%	2–3	[51]
Support Vector Machine (SVM)	Prediction of EPA content in <i>Nannochloropsis gaditana</i>	Growth conditions	EPA content	93%	2–3	[52]
Random Forest (RF)	Prediction of lipid content in <i>Tetraselmis suecica</i>	Growth conditions	Lipid content	94%	2–3	[53]
Artificial Neural Network (ANN)	Prediction of chlorophyll content in <i>Ankistrodesmus falcatus</i>	Nutrient levels	Chlorophyll content	95%	2–3	[54]

such as dissolved oxygen levels, biomass density, and nutrient depletion, using sensors integrated into the cultivation setup. Real time monitoring platforms that combine IoT sensors with ML algorithms are illustrated schematically in Fig. 1, showing how sensor data can be integrated into predictive models for process monitoring and operational control. Through autonomous adjustments of cultivation conditions, these systems prevent operational disruptions and optimise overall performance [55].

Monitoring requirements in microalgae systems are strongly influenced by cultivation scale and reactor configuration. Open raceway ponds typically rely on a limited number of robust sensors for parameters such as pH, temperature, and dissolved oxygen due to environmental variability and lower instrumentation density. In contrast, closed photobioreactors enable higher-resolution monitoring through integrated sensors for parameters including light intensity, gas exchange, and nutrient dynamics. At laboratory scale, high-frequency

research-grade sensors allow detailed data acquisition for model development, whereas pilot-scale systems require cost-effective and robust sensors with reliable wireless or automated data transmission. Commercial facilities typically require GMP-compliant instrumentation with automated data logging and traceability to meet regulatory and operational requirements. These differences in sensor deployment and data quality strongly influence the reliability and scalability of AI-driven monitoring and control strategies.

A key strength of AI systems lies in their ability to anticipate and manage both operational and environmental challenges in microalgae cultivation. By analysing historical data alongside weather patterns and operational trends, AI can predict potential issues such as temperature fluctuations, pH imbalances, or equipment malfunctions. Systems that implement foresight-driven preventive actions enable stable and reliable cultivation processes, minimising the impact of unexpected events on overall productivity [57]. The integration of ML into microalgae cultivation has thus established a data-centric approach that optimises real-time operations, enhances system resilience, and supports the development of sustainable microalgae-based bioprocessing technologies.

The development of microalgae-derived products has progressed significantly through the implementation of AI technologies, with machine learning playing a central role. A new generation of investigations uses advanced computing systems to discover novel product development ways and product enhancement opportunities. Several microalgae species have already been evaluated using AI methods,

particularly machine learning to identify or enhance commercially desirable traits such as high protein content, lipid accumulation, pigment production, or tolerance to wastewater conditions (Table 2). Table 2 summarises microalgae species and their AI-identified traits for potential commercial applications. The table also indicates approximate technology readiness levels (TRLs) to provide an overview of the technological maturity of the reported studies. As summarised in Table 2, *Chlorella* and *Nannochloropsis* are most common due to their high baseline protein/lipid content, but ML has focused more on lipid/pigment enhancement than protein optimisation or wastewater tolerance; a key gap is the lack of food-safety validation for AI-selected traits, limiting direct application to alternative foods. The application of AI establishes strain selection and performs genetic engineering as significant steps in developing microalgae products. A typical workflow for AI guided strain selection and genetic engineering of microalgae strains is presented in Fig. 2, outlining the sequence from biological data acquisition and model training to trait prediction and strain optimisation. Big datasets include microalgae genomic information and metabolic pathways, and phenotypic characteristic data help artificial intelligence systems to find appropriate microalgae strains that develop desired qualities such as higher biomass quantity or better lipid content, or stress resistance capability. Through strain selection and genetic engineering, the generation of unique microalgae strains becomes possible and achieves optimization for specific product production, including biofuels, nutraceuticals, or cosmetic compounds [72]. However, many AI assisted

Table 2 Microalgae species and their AI-identified traits for commercial applications with approximate technology readiness levels (TRLs)

Microalgal species	AI-identified traits	Input data	ML method used	Commercial application	TRL	References
<i>Chlorella vulgaris</i>	High protein content	Nutrient levels	ANN	Food additives, Dietary supplements	3–4	[58]
<i>Arthrospira platensis</i>	Fatty acid and vitamin richness	Growth conditions	SVM	Bioenergy, Health supplements	3–4	[59]
<i>Haematococcus pluvialis</i>	High astaxanthin production	Growth conditions	RF	Aquaculture feed	4–5	[60]
<i>Nannochloropsis gaditana</i>	High EPA content	Growth conditions	GA	Omega-3 supplements	3–4	[61]
<i>Dunaliella salina</i>	High beta-carotene	Growth parameters	DT	Dietary supplements	4–5	[62]
<i>Isochrysis galbana</i>	High fucoxanthin	Extraction parameters	ANN	Cosmetics	3–4	[63]
<i>Skeletonema costatum</i>	High lipid content	Growth conditions	SVM	Biofuel production	2–3	[50]
<i>Auxenochlorella protothecoides</i>	High lipid content	Growth conditions	ANN	Wastewater treatment, biodiesel production	3–4	[64]
<i>Tetraselmis chuii</i>	Astaxanthin production	Growth conditions	RF	Antioxidant Supplements	3–4	[65]
<i>Porphyridium cruentum</i>	High polysaccharide content	Growth parameters	GA	Pharmaceuticals	3–4	[66]
<i>Scenedesmus obliquus</i>	High biomass	Growth conditions	ANN	Wastewater treatment	3–4	[67]
<i>Chlamydomonas reinhardtii</i>	High hydrogen gas production	Growth parameters	SVM	Renewable energy	2–3	[68]
<i>Phaeodactylum tricornutum</i>	High EPA content	Growth parameters	RF	Nutraceuticals	3–4	[69]
<i>Botryococcus braunii</i>	High hydrocarbon content	Nitrogen limitations and light availability	GA	Biofuel production	2–3	[70]
<i>Ankistrodesmus falcatus</i>	High chlorophyll levels	Growth conditions	ANN	Dietary supplements, food colourants	3–4	[71]

Fig. 2 Schematic representation of IoT enabled monitoring and machine learning based control in microalgae cultivation systems

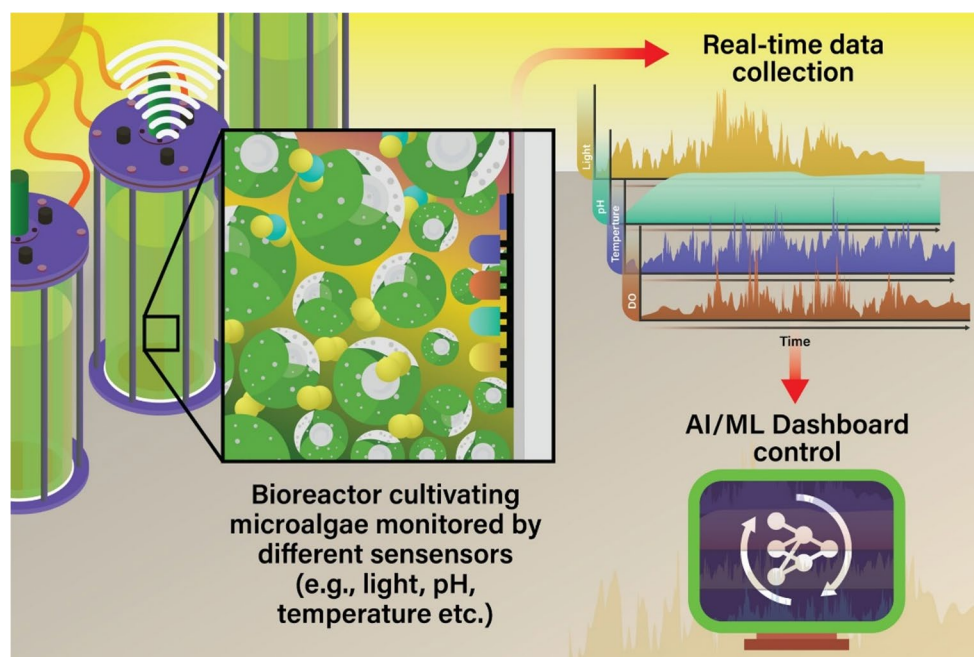
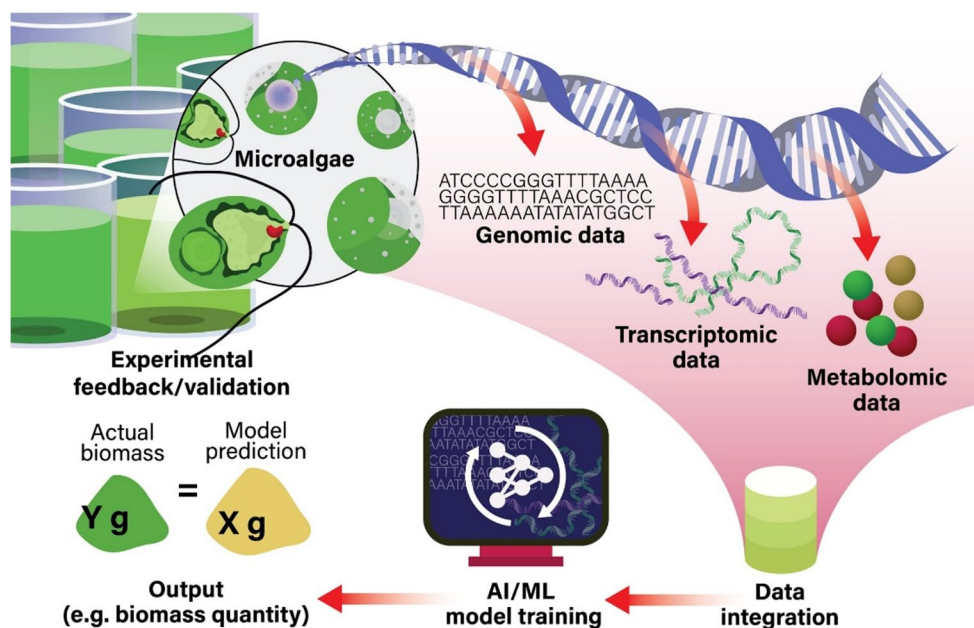


Fig. 3 Workflow for AI guided strain selection and genetic engineering of microalgae



strain engineering approaches remain at an early stage of technological development and typically correspond to low technology readiness levels (TRL 1–3). Most reported studies rely on computational screening, laboratory scale validation, or proof of concept demonstrations rather than large scale cultivation trials. As a result, their direct application to near term food production or wastewater integrated microalgae systems remains limited, and substantial experimental validation and regulatory assessment will be required before such approaches can be implemented in industrial environments.

Artificial intelligence technologies, including machine learning, enable better separation methods that help to generate better microalgae products, together with higher yields. Combining artificial intelligence and machine learning technologies, data analysis finds the ideal conditions that lead to high-value compound extractions from microalgae biomass, such as carotenoids or omega-3 fatty acids. AI-driven process control systems enable engineers to maximize purification methods such that consistent products with reduced waste arise [73]. For these models to be useful at an industrial scale, rigorous validation is

essential. Best practice includes 70/15/15 train/validation/test splits, k-fold cross-validation, and most importantly, external validation on independent pilot or full-scale datasets with different strains, reactor geometries, or wastewater compositions. Transfer learning (pre-training on large public datasets and fine-tuning) and domain-adaptation techniques are increasingly used to improve robustness when moving from lab to outdoor or wastewater conditions Saha, Chauhan (60).

Predictive analytics combined with pattern recognition techniques helps researchers to find fresh uses for microalgae components while looking at mixing possibilities and creating tailored formulations addressing current market needs. Supporting the growth and development of the microalgae bioprocessing sector, data-driven product optimization strategies increase the practicality and adaptability of microalgae-based products. One important progress in this field is the combination of artificial intelligence and machine learning technology in microalgae product development and optimization. These sophisticated computing techniques enable research teams to find new opportunities and reach higher yields coupled with better quality standards, therefore driving the development of sustainable microalgae-based products [74].

Applications of ML in Microalgae Bioprocesses for Alternative Foods and Water Sustainability

While Sect. 3 described the main ML tasks and modeling approaches, this section highlights concrete, application-driven examples that demonstrate how these tools are already delivering real-world impact in alternative-protein production and water-treatment systems.

ML for Alternative Food Products and Sensory Quality

Several studies have used AI to develop microalgae strains and processes specifically tailored for food applications. For instance, machine-learning-guided screening of large culture collections has identified *Chlorella* and *Spirulina* variants with higher protein content (up to 65–70% dry weight) and improved amino-acid profiles suitable for meat/dairy alternatives [5]. CNN-based image analysis and random-forest models have been trained to predict and minimise off-flavour compounds (geosmin, 2-methylisoborneol) and chlorophyll-related green colour by optimising light spectrum, harvesting time, and mild bleaching conditions without non-GRAS reagents [38, 45]. In one promising case, ANN models successfully optimized encapsulation

parameters for *Spirulina* and *Chlorella* proteins, improving stability and masking undesirable taste in plant-based beverages and extruded products.

Beyond strain selection and protein content prediction, AI models have been extensively applied to optimise the techno-functional properties of microalgae proteins essential for food formulation. Artificial neural networks trained on extraction parameters (pH, temperature, enzyme concentration, sonication time) have successfully predicted the emulsification activity and foaming capacity of *Chlorella vulgaris* protein isolates, achieving correlation coefficients above 0.92 between predicted and measured values. These models identified optimal extraction conditions (pH 8.5, 40 °C, 30 min enzymatic hydrolysis with Alcalase) that produced isolates with 91.5% foaming capacity and 75.1% emulsion stability, values exceeding those of soy protein isolate under identical testing conditions [75]. Similarly, genetic algorithms coupled with response surface methodology have optimised ultrasound-assisted extraction of *Arthrospira platensis* proteins, increasing solubility from 62% to 89% at neutral pH while preserving essential amino acid profiles [76].

Consumer acceptance of microalgae-based foods remains constrained by characteristic off-flavours and intense green colouration. Recent advances employ convolutional neural networks (CNNs) trained on hyperspectral imaging data to predict chlorophyll and carotenoid concentrations in real-time during cultivation, enabling dynamic light adjustment that reduces pigmentation by up to 40% without compromising biomass yield [56]. For off-flavour mitigation, random forest models trained on gas chromatography-mass spectrometry datasets have identified critical control points for geosmin and 2-methylisoborneol accumulation. These models revealed that maintaining cultivation temperature below 22 °C and implementing specific light-dark cycles (16:8 h) reduces geosmin concentrations by 65% compared to standard conditions [49].

Food safety prediction represents an emerging application of machine learning in microalgae processing. Support vector machine models have been applied to predict heavy metal biosorption and accumulation in algal systems using physicochemical parameters of the growth medium, achieving prediction accuracies exceeding 90% in several studies [77]. Such models provide a potential framework for predicting contaminant transfer risks in biomass cultivated using wastewater-derived nutrients and could support early interventions such as adjusting cultivation conditions or harvesting strategies. In addition, machine learning classifiers such as random forests have been widely applied in food safety risk assessment to detect microbial contamination based on processing conditions and environmental variables, achieving classification accuracies between 90 and

96% in pathogen prediction studies [78]. These approaches provide a foundation for future AI-assisted safety monitoring and quality assurance in microalgae-based food production systems.

ML Applications in Wastewater-coupled Microalgae Systems for Water Sustainability and Nutrient Recovery

One of the most promising routes for sustainable microalgae cultivation is direct integration with municipal or industrial wastewater treatment, where microalgae (usually in consortia with bacteria) use the wastewater nutrients while simultaneously cleaning the water. ML tools are now routinely used to maximise both biomass/protein yield and pollutant-removal performance in these systems. Support-vector regression and random-forest models have successfully predicted nitrogen and phosphorus uptake rates, allowing dynamic adjustment of hydraulic retention time, light intensity, or inoculation ratios to achieve >90% nutrient removal while producing food-grade or feed-grade biomass [28, 57], Hossain, Sultana [79]. In industrial effluents (paper-mill, dairy, or food-processing wastewater), hybrid physics-ML frameworks forecast granule formation, settling behaviour, and heavy-metal or micropollutant uptake, reducing or eliminating chemical flocculants that would otherwise make the biomass unsuitable for food applications [16, 29].

ML has also been applied to optimise CO₂ bio-fixation and emergent pollutant removal in wastewater-fed photobioreactors. For example, ANN and reinforcement-learning controllers adjust aeration and CO₂ sparging in real time to maximise carbon capture (up to 1.83 kg CO₂ per kg biomass) while maintaining high ammonium and phosphate removal [47]. Digital-twin platforms that combine CFD with ML correction have been demonstrated at pilot scale for tertiary treatment, predicting long-term performance under fluctuating influent loads and enabling true closed-loop operation that recycles treated water and nutrients back into the cultivation loop.

Machine learning models have substantially improved the prediction and control of nutrient removal dynamics in wastewater-coupled systems. Support vector regression models trained on influent characteristics (chemical oxygen demand, total nitrogen, total phosphorus) and operational parameters (hydraulic retention time, temperature, light intensity) have achieved >92% accuracy in predicting effluent quality across municipal, agricultural, and industrial wastewater types [57]. In paper pulp wastewater treatment, artificial neural networks incorporating real-time sensor data (pH, dissolved oxygen, ammonium concentration) enabled dynamic adjustment of hydraulic retention time from 6 to 10 days, maintaining >95% nitrogen removal efficiency

despite influent fluctuations of $\pm 30\%$. Random forest models applied to dairy wastewater identified light intensity ($180\text{--}200\text{ }\mu\text{mol m}^{-2}\text{ s}^{-1}$) and algal-bacterial inoculation ratio (3:1) as critical parameters for achieving simultaneous >90% phosphorus removal and biomass productivity exceeding $0.8\text{ g L}^{-1}\text{ day}^{-1}$ [29].

Contaminant monitoring and prediction have advanced through machine learning applications that address food safety concerns for wastewater-grown biomass. Support vector machines trained on wastewater composition data and cultivation parameters predicted cadmium and lead accumulation in *Chlorella* biomass with 93% accuracy, enabling identification of safe harvesting windows before contaminants exceed regulatory limits [12]. For pathogen surveillance, random forest models incorporating 16 S rRNA sequencing data and physicochemical parameters achieved 91% sensitivity in predicting *Salmonella* presence in algal-bacterial consortia, allowing targeted interventions such as increased aeration or pH adjustment [38]. Emerging pollutant removal, including pharmaceutical residues (ibuprofen, diclofenac) and microplastics, has been modelled using artificial neural networks that achieved 87–92% accuracy in predicting degradation rates, enabling real-time adjustment of hydraulic retention time to ensure complete removal before harvest [80].

Biomass quality prediction for food applications represents a critical integration point between wastewater treatment and protein production. Random forest models trained on influent C: N ratio, ammonium concentration, and harvesting time have predicted protein accumulation in municipal wastewater-cultivated *Chlorella vulgaris* with 92% accuracy, identifying conditions that consistently produce biomass exceeding 50% protein content, the threshold for economic viability in feed and food applications [4, 10]. Artificial neural networks have been applied to predict essential amino acid profiles of biomass cultivated on agricultural runoff, achieving 89% accuracy in forecasting whether harvested material meets FAO reference patterns for human nutrition [81]. These predictive capabilities enable operators to manage wastewater treatment systems with explicit targets for biomass quality, creating a credible pathway for food-grade production from non-conventional water sources.

ML for Industrial-Scale Operation and Digital Twins

Scaling microalgae processes to commercial food production requires robust, fault-tolerant operation. Digital-twin platforms that mirror full-scale raceway ponds or photobioreactors in real time have been built using hybrid physics-ANN architectures. These twins predict contamination events, pH swings, or temperature excursions days

in advance and automatically adjust aeration or CO₂ dosing [38, 56]. Predictive-maintenance models trained on sensor data (vibration, power draw, optical density trends) have reduced unplanned downtime by 30–50% in large outdoor facilities. In one notable industrial case, a reinforcement-learning controller optimised harvesting frequency and dewatering in a 10,000 m² raceway pond producing *Spirulina* for food supplements, cutting energy consumption by 25% while maintaining strict microbiological quality [55]. ML methods for optimising harvesting, drying, and extraction steps are summarised in Table 3, with a focus on models that balance efficiency with food-grade constraints. Table 3 also indicates approximate technology readiness levels (TRLs) to provide an overview of the technological maturity of the reported approaches. Table 3 highlights the growing but limited use of ML in downstream process chains and wastewater systems, with ANN and genetic algorithms most common for extraction yield or moisture optimisation but hybrid models still rare; notable gaps include underuse of interpretable models, reliance on small lab datasets, and poor external validation across varying wastewater types or scales, which must be addressed for reliable industrial food and water applications.

ML-Enabled Integration of Food Production with Wastewater Treatment

This subsection presents an integrated framework where these applications converge, enabling the safe transition from nutrient-rich wastewater to food-grade protein ingredients. In the cultivation stage, microalgae grown in wastewater must simultaneously achieve high biomass productivity, efficient nutrient removal, and minimal contaminant uptake. Artificial neural networks and random forest models predict optimal parameters (hydraulic retention time, light intensity, CO₂ supplementation) that balance biomass productivity with nitrogen and phosphorus removal exceeding 90%, while support vector machines trained on wastewater composition predict heavy metal accumulation with >90% accuracy to identify safe harvesting windows before contaminants approach food safety thresholds [16, 47].

Before biomass can proceed to food processing, rigorous safety screening is essential, where machine learning classifiers enable rapid prediction of heavy metals (EU thresholds), pathogens such as *Salmonella* and *Listeria*, and emerging pollutants, facilitating immediate diversion of non-compliant batches to non-food applications such as biofuels or fertilisers. During processing, validated biomass undergoes cell disruption and protein extraction, with genetic algorithms and Bayesian optimisation determining optimal parameters that maximise protein yield while preserving techno-functional properties [64]. Processing

Table 3 Comparative summary of harvesting and drying methods for microalgae optimized using machine learning approaches with approximate technology readiness levels (TRLs)

Methods	ML Technique	Key Findings	Target Species	TRL	References
Electroflo-tation	Optimization algorithms	96.2% higher harvesting efficiency for <i>Tribonema sp.</i>	<i>Tribonema sp.</i> , <i>Scenedesmus sp.</i> , <i>Pandorina sp.</i>	4–5	[37]
Jameson cell	Process modelling	Energy efficient for microalgal harvesting	Various microalgal species	3–4	[82]
Electrocoagulation	Design optimization	96.18% harvesting efficiency for <i>Tetraselmis sp.</i>	<i>Tetraselmis sp.</i>	4–5	[83]
Combined flocculants	Experimental optimization	Enhanced harvesting efficiency	<i>Chlorella vulgaris</i>	3–4	[84]
Spray drying	Process stimulation	Suitable for high-value products	Different microalgal species	6–7	[85]
Freeze drying	Process modelling	Preserve cell constituents	Various microalgal species	5–6	[86]
Drum drying	Energy optimisation	Simultaneous sterilisation and cell rupturing	<i>Spirulina platensis</i>	5–6	[87]
Solar drying	Biomass quality assessment	Economical	Various microalgal species	4–5	[88]
Sand-Enhanced electro flocculation	Experimental optimisation	Cost-effective harvesting	<i>Dunaliella salina</i>	3–4	[89]
Tray drying	Response surface methodology	Energy efficient	<i>Chlorella vulgaris</i>	4–5	[90]
Membrane filtration	Process modeling	Low energy consumption	Various microalgal species	5–6	[91]
Cross-flow filtration	Process modelling	Suitable for large-scale operations	Different microalgal species	5–6	[92]
Decanter centrifugation	Process optimisation	Effective for continuous operations	Different microalgal species	6–7	[93]
Organic dlocculation	Experimental optimisation	Environment friendly	Various microalgal species	4–5	[94]
Diatomite dynamic membrane	Experimental optimisation	Cost-effective	<i>Chlorella pyrenoidosa</i>	3–4	[95]

serves a dual safety function as AI models predict contaminant partitioning behaviour, showing that heavy metals and lipophilic pollutants concentrate in lipid or carbohydrate fractions, reducing their levels in final protein isolates by 60–80%, while decolorization and deodorization steps are optimised through machine learning to minimise off-flavours using food-grade reagents.

In the application stage, resulting protein ingredients are incorporated into formulated food products through AI-assisted formulation design that optimises ingredient blends for sensory profiles and nutritional targets, while treated water from cultivation meeting discharge standards is recycled for irrigation or industrial processes, closing the water loop. Digital twin platforms integrate process data with AI predictions to enable continuous monitoring of both product quality and water quality, ensuring system-wide operation within safety and sustainability boundaries. This integrated framework demonstrates that AI and machine learning are essential enablers of a circular bioeconomy where wastewater treatment and food production are synergistically combined, with the same models (as described in Sect. 5.2) that predict nutrient removal informing food safety decisions and the same sensors that monitor water quality guiding harvesting windows. Section 5 addresses the challenges in achieving such integration, including data availability, model interpretability, and regulatory acceptance. The comparative analysis of machine learning models used in microalgae systems reveals distinct strengths and limitations depending on the application context. Table 4 summarises the key characteristics of the most used ML approaches, including their advantages, limitations, and suitability for food and water sustainability applications. As shown in Table 4, while artificial neural networks dominate predictive modelling tasks due to their ability to capture nonlinear biological relationships, interpretable models and hybrid approaches may be more suitable for applications requiring transparency and regulatory compliance.

Challenges and Limitations

Technical and Data Challenges

In microalgae bioprocessing, the whole use of artificial intelligence particularly machine learning depends on thorough and high-quality datasets for correct operation. These datasets are a basic base for training and model refining for the predictive models for optimisation and decision-making. Microalgae with artificial intelligence require operational data alongside process information, environmental data, and market conditions, as well as data on cultivation parameters and strain attributes [96].

Environmental parameters like temperature, light intensity, pH, dissolved oxygen, and nutrient concentrations provide the training basis for AI models to identify optimal growth conditions. However, operational data for microalgae-bacteria wastewater systems are often fragmented, lacking integrated information on biomass composition, granule properties, and environmental drivers, which significantly restricts robust ML model development [28]. Even detailed studies on photogranular consortia that report pigments, nitrogen species, and biomass yield under different inoculation strategies, usually present the data in a format tailored to process engineering rather than to model training, which limits their direct reuse for developing robust ML models [16]. These models learn strain patterns and trait relationships between genetic markers, metabolic pathways, and desired characteristics like high biomass and lipid content through genomic, metabolomic, and phenotypic analysis [97]. The microalgae-based pipeline optimization depends on historical records analysis, which enables artificial intelligence systems to reveal unobvious connections leading to maximum bioprocessing performance [98].

Supervised and unsupervised learning algorithms within machine learning consume vast datasets to discover ideal cultivation parameters and choose superior microalgae strains, together with improved production methods and product advancements. Genomic data linked with cultivation performance allows researchers to identify genetic markers for high biomass and lipid production, which becomes the basis for microalgae strain genetic engineering strategies [56]. The advancement of microalgae bioprocessing depends upon the continuous growth of powerful and numerous datasets. Researchers and industry professionals must invest in microalgae database collection alongside curation activities to maximize the use of AI for developing sustainable, innovative microalgae-based applications through operational enhancement.

Although mechanistic or kinetic models shine in this regard but typically fall short in prediction accuracy due to oversimplified representations. ML models lack natural awareness of the underlying physics of algae cultivation processes. Combining the strengths of ML and physics-based models in a hybrid learning style presents a potential fix and helps to minimise the restrictions of each, therefore producing more accurate forecasts, better extrapolation capacity, more interpretability, and less data needed in modelling. Gaussian filters, among other filtering methods, can help to significantly reduce data noise and mistakes. Moreover, addressing missing values is crucial; methods such as imputation with statistical metrics like mean, median, or through more sophisticated techniques defined in provide the overview of simple imputations, regression imputations, hot-deck imputations, expectation-maximization, multiple

Table 4 Comparative evaluation of machine learning model types used in microalgae bioprocess optimisation for food and water sustainability applications

Model type	Strengths	Limitations	Typical applications	Suitability for Food/Water
Artificial Neural Networks (ANN)	Models non-linear relationships; high accuracy with large datasets; flexible architecture	Black-box; requires large training data; prone to overfitting; limited extrapolation	Biomass prediction; protein/lipid content forecasting; yield optimisation	High for prediction tasks; limited for regulatory compliance due to interpretability issues
Support Vector Machines (SVM)	Effective with high-dimensional data; robust with small datasets; good classification performance	Less effective with noisy data; parameter sensitivity; limited probabilistic outputs	Species classification; contaminant detection; quality grading	High for classification tasks where interpretability is balanced with accuracy
Random Forests (RF)	Feature importance ranking; handles mixed data types; resistant to overfitting	Less accurate than boosted methods; can be computationally intensive	Growth rate prediction; nutrient removal forecasting; feature selection	High for identifying key process parameters and guiding experimental design
Genetic Algorithms (GA)	Excellent for multi-objective optimisation; searches large parameter spaces	Requires integration with predictive models; computationally expensive	Optimisation of cultivation conditions; parameter tuning; process design	High for process optimisation when combined with interpretable models
Decision Trees (DT)	Highly interpretable; simple implementation; transparent decision rules	Instability; tendency to overfit; lower predictive accuracy	Quality control; regulatory documentation; simple classification	High for applications requiring transparency and audit trails
Convolutional Neural Networks (CNN)	Excellent for image and spectral data; automatic feature extraction	Requires large datasets; computationally intensive; black-box	Species identification from microscopy; pigment prediction from spectra	Moderate for food quality monitoring; requires validation for regulatory use
Hybrid/Physics-Informed Models	Combines ML with mechanistic understanding; improved extrapolation; regulatory acceptability	Complex implementation; requires domain expertise	Wastewater granule dynamics; contaminant fate prediction; scale-up modelling	High for integrated food-water systems requiring regulatory approval

imputations, and imputation methods inspired by ML to handle the missing data value in datasets. Data augmentation via GANs and autoencoders is useful, but synthetic data must accurately reflect real features to avoid biases that compromise model performance. Developing robust AI models for the integrated pathway (as described in Sect. 4.4) requires datasets that simultaneously capture influent wastewater composition, cultivation dynamics, contaminant concentrations, and final product quality. Currently, such integrated datasets are rare; most studies report either wastewater treatment performance or biomass composition, but not in a format amenable to machine learning. Future efforts should prioritise open-access, multi-domain databases to enable training of hybrid models that can predict final food safety from upstream conditions.

Regulatory and Safety Challenges

Regulatory and safety aspects represent a major hurdle, particularly for food-grade products. Novel food regulations (e.g., EU Regulation 2015/2283 or FDA GRAS requirements) demand extensive validation of any AI-optimised

strain or process to prove safety, including absence of toxins, allergens, or contaminants introduced during cultivation or downstream steps [5, 15]. For wastewater-integrated systems, additional scrutiny is needed to ensure no transfer of heavy metals, pathogens, or emergent pollutants into the biomass, which could otherwise preclude water reuse or food-chain entry. AI models themselves must comply with good manufacturing practices (GMP/GxP), meaning their outputs (e.g., optimised growth conditions) require traceable documentation and reproducibility for regulatory audits.

Regulatory approval represents a critical determinant of commercial viability for microalgae derived food ingredients. In the European Union, foods not consumed to a significant degree before 15 May 1997 are classified as novel foods under Regulation (EU) 2015/2283 and require comprehensive safety assessment including compositional analysis, toxicological evaluation, and verification of production processes. Importantly, novel food status is species specific: while *Arthrospira platensis* and *Chlorella vulgaris* whole biomass have an established history of safe consumption, protein isolates from these species and products derived

from novel microalgal taxa may trigger full authorisation requirements. The US FDA GRAS framework similarly requires demonstration of safety under intended conditions of use [58]. These regulatory requirements become substantially more stringent when wastewater derived nutrients are used in cultivation systems.

Authorities require clear evidence that heavy metals (EU guidance values: $Pb \leq 3$ mg/kg, $Cd \leq 1$ mg/kg, $Hg \leq 0.1$ mg/kg), pathogens (*Salmonella* absent in 25 g), and emerging contaminants are not transferred to the biomass [99]. Consequently, most commercial microalgae-based food products are currently produced using defined growth media rather than wastewater streams, with the integrated wastewater to food pathway remaining at pilot demonstration stage. AI enabled monitoring can support regulatory compliance through predictive models that forecast contaminant accumulation with $>90\%$ accuracy, enabling proactive harvesting before thresholds are exceeded, and through machine learning classifiers that generate digital audit trails for regulatory inspection. However, regulatory agencies currently lack established frameworks for validating AI models used in food safety decisions, favouring interpretable hybrid approaches over purely black box methods. This intersection of AI optimisation with regulatory compliance remains an active area of development.

Explainability and Trust in AI Models

Explainability and trust in AI models are equally critical, especially in safety-sensitive domains like food production and wastewater treatment. Many high-performing “black-box” models (e.g., deep neural networks) provide accurate predictions but offer little insight into why a certain nutrient level or harvesting time was recommended, which erodes operator confidence and hinders regulatory approval [38, 100]. To mitigate this, explainable AI techniques such as feature-importance analysis (SHAP or LIME values), decision-tree surrogates, or physics-informed constraints that highlight key variables like pH or light intensity are essential for transparent decision-making. In wastewater applications, these techniques can clarify how model recommendations affect nutrient removal or granule stability, building trust among treatment-plant operators [16, 28].

Ethical, Environmental, and Social Challenges

Ethical, environmental, and social considerations also warrant attention. The computational demands of training complex ML models contribute to high energy use and carbon emissions, potentially offsetting the environmental benefits of CO_2 -sequestering microalgae [13]. Data governance issues arise when proprietary datasets from industrial

wastewater streams or strain libraries are used, raising questions about intellectual property, privacy (if sensor data includes location specifics), and equitable access for smaller enterprises in developing regions. Socially, consumer acceptance of AI-designed microalgae foods remains uncertain; consumers may resist “engineered” proteins due to perceptions of unnaturalness, particularly if green colour or off-flavours persist despite optimisation [5]. Broader ethical concerns include ensuring that AI-driven efficiencies do not exacerbate social inequalities, such as by displacing manual labour in rural cultivation facilities without adequate retraining.

Practical Barriers to Adoption

Practical barriers to adoption cannot be ignored. Many microalgae producers, especially small and medium-sized enterprises, lack the interdisciplinary skills (data science + phycology) needed to implement AI tools, especially those based on machine learning, along with the sensor infrastructure (IoT networks, high-resolution spectrometers) required for real-time data collection [20, 38]. Initial digitalisation costs can be prohibitive, and cultural resistance in traditional industries where empirical methods have long dominated may slow uptake. In wastewater contexts, variability in influent quality (seasonal or source-dependent) further complicates model training and deployment.

Beyond the practical challenges of digitalisation and data infrastructure, it is also important to recognise that many proposed wastewater–microalgae systems remain at early stages of technological development. Despite the considerable potential of coupling microalgae cultivation with wastewater treatment for sustainable protein production, most systems reported in the literature remain at laboratory or pilot scale, with relatively few examples approaching commercial implementation. Pilot-scale studies have demonstrated effective nutrient removal and algal biomass production in municipal and agricultural wastewater streams; however, these systems have primarily been developed for bioenergy, fertiliser, or environmental remediation applications rather than food production. The use of wastewater-derived nutrients for food-grade microalgae cultivation introduces additional safety challenges, including the need for strict monitoring of heavy metals, organic contaminants, pathogens, and toxin accumulation. Compliance with regulatory frameworks such as EU Novel Food authorisation and FDA GRAS requirements therefore requires extensive validation before wastewater-grown biomass can enter the food chain. Consequently, most commercially available microalgae-based food products are currently produced in controlled cultivation systems using defined growth media rather than wastewater streams. Overall, while

wastewater-based microalgae systems show strong promise for sustainable nutrient recovery, the integrated pathway linking wastewater treatment with food-grade protein production remains largely at the pilot demonstration stage.

Addressing these barriers will require coordinated advances in data infrastructure, process engineering, and regulatory validation. In practical terms, the development of standardized datasets from pilot and industrial cultivation systems will be essential for training robust AI models capable of handling environmental variability. Similarly, wider deployment of modular sensor platforms and interoperable data pipelines could reduce the digitalisation burden for smaller producers and enable more reliable real-time monitoring. From a regulatory perspective, clearer validation frameworks for AI-assisted cultivation and wastewater-based nutrient recovery will be necessary to support safe integration into food production systems. Progress in these areas would significantly improve the feasibility of AI-enabled microalgae platforms and help bridge the current gap between laboratory-scale innovation and industrial deployment.

Conclusion and Prospects

This review has demonstrated that AI tools, particularly machine learning, can already deliver substantial improvements in microalgae bioprocessing for alternative proteins and water sustainability. Key advances include marked enhancements in biomass productivity through predictive growth modelling, high-efficiency nitrogen and phosphorus removal in wastewater-integrated cultivation, and meaningful reductions in operational energy demand when real-time process control is applied. In the near future, hybrid physics-AI frameworks and digital-twin platforms are expected to evolve from laboratory-confined modelling toward industrial-scale deployment, enabling more stable and resource-efficient operational pipelines that support circular-economy integration.

The production of microalgae-derived proteins for alternative foods, water reuse systems, and sustainable feed applications continues to face persistent constraints. These include the need for rapid identification of high-protein and contamination-safe strains, reduction in cultivation and downstream processing costs, mitigation of intense green coloration in final ingredients, improvement in organoleptic properties, and strict compliance with food-grade microbial safety, ingredient purity, toxin-free biomass validation, and reproducibility of optimised workflows. Although multi-omics platforms, including genomics, transcriptomics, proteomics, and metabolomics, have the biological depth to map system-wide routes influencing both protein quality

and quantity, these technologies inherently generate large, heterogeneous datasets that exceed the processing capacity of traditional analytical pipelines. AI and computational systems are, therefore, essential not as substitutes for experimentation, but as integrative engines that can unify fragmented molecular, process, environmental, wastewater, and production datasets into interpretable learning patterns. Such patterns can guide targeted experimental design, optimise cultivation steps, refine harvesting windows, and inform product-ingredient strategies without compromising biological feasibility or industrial compliance.

The realistic potential of AI for future food-water bioprocessing chains will rely on continuous expansion of curated, domain-aligned production datasets that combine strain-level multi-omics profiles with cultivation and wastewater-data indicators, while embedding model explainability and audit-level traceability for regulator and operator confidence. Furthermore, digital-twin systems must progress from lab-scale simulations to validated, real bioreactor and open-pond environments, incorporating water- and influent-driven variability, climate-linked stress parameters, process control feedback loops, and contamination-safe harvesting and extraction strategies that respect food-grade coagulant, solvent, and processing boundaries. Finally, pilot-to-industrial transition systems must validate realistic performance thresholds, benchmark AI performance against sustainability and techno-economic viability targets and define failure-mode limits under real production setups.

A realistic assessment of current technological maturity highlights the significant gap between laboratory innovation and industrial deployment. Many studies reporting high biomass productivity or process optimisation are conducted in small laboratory photobioreactors or controlled synthetic media, which do not fully capture the environmental variability, contamination risks, and economic constraints encountered in large-scale systems. In practice, microalgae technologies span multiple Technology Readiness Levels (TRLs). Fundamental strain development and AI-driven optimisation models typically operate at TRL 1–3, pilot-scale cultivation and wastewater treatment demonstrations correspond to TRL 4–6, while commercially established systems, such as *Spirulina* and *Chlorella* production in raceway ponds or closed photobioreactors, represent TRL 8–9. However, the fully integrated pathway combining wastewater treatment with food-grade microalgae protein production remains largely at the pilot demonstration stage, requiring further validation of contaminant safety, regulatory approval pathways, and long-term operational stability before widespread commercial adoption can be achieved. Microalgae biotechnology now stands at a multidisciplinary intersection of alternative protein systems, water circularity, and computational biology. The near-term future will depend upon

model-experimental integration, systems-level data fusion, regulatory-safe AI workflows, and field-validated digital infrastructures to translate predictive capability into scalable, sustainable, and food-chain-safe protein innovation.

Key References

- Coşgun A, Günay ME, Yıldırım R. Machine learning for algal biofuels: a critical review and perspective for the future. *Green Chemistry*. 2023;25(9):3354-73.
 - This article provides a specialized analysis of machine learning applications for algal biofuel production, critically evaluating data-driven strategies to enhance lipid yield and process economics within sustainable chemistry frameworks.
- Helmy M, Elhalis H, Liu Y, Chow Y, Selvarajoo K. Perspective: multiomics and machine learning help unleash the alternative food potential of microalgae. *Advances in Nutrition*. 2023;14(1):1-11.
 - This perspective highlights the convergence of multiomics data and machine learning to optimize microalgae as a sustainable source of nutritional proteins and functional food ingredients, expanding the bioeconomy beyond biofuels.
- Liang C, Yang Y, Xia Y, Yuan W, Chen J, Zheng Z, et al. The optimization of *Chlorella vulgaris* flocculation harvesting by chitosan and calcium hydroxide. *Indian journal of microbiology*. 2022;62(2):266-72.
 - This review details the integration of Internet of Things (IoT) sensors and networks with microalgae cultivation, outlining a practical pathway towards fully automated, data-driven smart farming systems for reliable biomass production.
- Long B, Fischer B, Zeng Y, Amerigian Z, Li Q, Bryant H, et al. Machine learning-informed and synthetic biology-enabled semi-continuous algal cultivation to unleash renewable fuel productivity. *Nature Communications*. 2022;13(1):541.
 - This high-impact study demonstrates a breakthrough application of machine learning coupled with synthetic biology to design a semi-continuous cultivation protocol, resulting in a transformative increase in algal biomass and fuel productivity.
- Oruganti RK, Biji AP, Lanuyanger T, Show PL, Sriariyanun M, Upadhyayula VK, et al. Artificial intelligence and machine learning tools for high-performance microalgal wastewater treatment and algal biorefinery: A critical review. *Science of The Total Environment*. 2023;876:162797.
 - This critical review synthesizes the application of ML for optimizing microalgae-based wastewater treatment and biorefinery operations, directly addressing the synergy between environmental remediation and the production of valuable biomass.
- Sátiro J, Fernandes JP, Mucha AP, Gomes AP, Calheiros CSC, Muñoz R, et al. Effect of slow feeding and hydraulic retention time on algal-bacterial aerobic granular sludge for low-nitrogen pulp mill wastewater treatment. *Journal of Environmental Chemical Engineering*. 2025:118216.
 - This study demonstrates how slow feeding strategies and hydraulic retention time critically influence the stability, nutrient removal performance, and granule formation of algal-bacterial aerobic granular sludge systems treating low-nitrogen pulp mill wastewater.
- Teng SY, Yew GY, Sukačová K, Show PL, Máša V, Chang J-S. Microalgae with artificial intelligence: A digitalized perspective on genetics, systems and products. *Biotechnology Advances*. 2020;44:107631.
 - This foundational review establishes the paradigm of integrating artificial intelligence across the entire microalgal value chain, from genetic engineering and cultivation systems to bioproduct development, providing a comprehensive digitalized framework for the field.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

- Erdaw MM. Contribution, prospects and trends of livestock production in sub-Saharan Africa: a review. *Int J Agric Sustain*. 2023;21(1):2247776.
- Gil M, Rudy M, Duma-Kocan P, Stanisławczyk R, Krajewska A, Dziki D, et al. Sustainability of Alternatives to Animal Protein Sources, a Comprehensive Review. *Sustainability*. 2024;16(17):7701.
- Toutirais L, Walrand S, Vaysse C. Are oilseeds a new alternative protein source for human nutrition? *Food Funct*. 2024;15(5):2366–80.
- Alexandratos N, Bruinsma J. World agriculture towards 2030/2050: the 2012 revision. 2012.
- Kumar R, Hegde AS, Sharma K, Parmar P, Srivatsan V. Microalgae as a sustainable source of edible proteins and bioactive peptides—current trends and future prospects. *Food Res Int*. 2022;157:111338.
- Salter AM, Lopez-Viso C. Role of novel protein sources in sustainably meeting future global requirements. *Proc Nutr Soc*. 2021;80(2):186–94.
- Matassa S, Batstone DJ, Hülsen T, Schnoor J, Verstraete W. Can direct conversion of used nitrogen to new feed and protein help feed the world? ACS Publications; 2015.
- Poore J, Nemecek T. Reducing food's environmental impacts through producers and consumers. *Science*. 2018;360(6392):987–92.
- Bleakley S, Hayes M. Algal proteins: extraction, application, and challenges concerning production. *Foods*. 2017;6(5):33.
- Sarker NK, Kaparaju P, Rao AR. Algal Bioremediation in Asian Developing Countries and Its Potential Applications. In: *Algae Mediated Bioremediation: Industrial Prospectives*, vol. 2. 2024. p. 715–35.
- Ahmed N, Ward JD, Thompson S, Saint CP, Diana JS. Blue-green water nexus in aquaculture for resilience to climate change. *Rev Fish Sci Aquac*. 2018;26(2):139–54.
- Chapman RL. Algae: the world's most important plants—an introduction. *Mitig Adapt Strat Glob Change*. 2013;18:5–12.
- Lim HR, Khoo KS, Chia WY, Chew KW, Ho S-H, Show PL. Smart microalgae farming with internet-of-things for sustainable agriculture. *Biotechnol Adv*. 2022;57:107931.
- Torres-Tiji Y, Fields FJ, Mayfield SP. Microalgae as a future food source. *Biotechnol Adv*. 2020;41:107536.
- Vieira MV, Pastrana LM, Fuciños P. Microalgae encapsulation systems for food, pharmaceutical and cosmetics applications. *Mar Drugs*. 2020;18(12):644.
- Satiro J, dos Santos Neto AG, Marinho T, Sales M, Marinho I, Kato MT, et al. The role of the microalgae–bacteria consortium in biomass formation and its application in wastewater treatment systems: a comprehensive review. *Appl Sci*. 2024;14(14):6083.
- Chew KW, Yap JY, Show PL, Suan NH, Juan JC, Ling TC, et al. Microalgae biorefinery: high value products perspectives. *Biore-sour Technol*. 2017;229:53–62.
- Oruganti RK, Biji AP, Lanuyanger T, Show PL, Sriariyanun M, Upadhyayula VK, et al. Artificial intelligence and machine learning tools for high-performance microalgal wastewater treatment and algal biorefinery: A critical review. *Sci Total Environ*. 2023;876:162797.
- Diankristanti PA, Ho NHE, Chen J-H, Nagarajan D, Chen C-Y, Hsieh Y-M et al. Unlocking the potential of microalgae as sustainable bioresources from up to downstream processing: A critical review. *Chem Eng J*. 2024:151124.
- Helmy M, Elhalis H, Liu Y, Chow Y, Selvarajoo K. Perspective: multiomics and machine learning help unleash the alternative food potential of microalgae. *Adv Nutr*. 2023;14(1):1–11.
- Dolganyuk V, Belova D, Babich O, Prosekov A, Ivanova S, Katsarov D, et al. Microalgae: A promising source of valuable bioproducts. *Biomolecules*. 2020;10(8):1153.
- Saratale RG, Ponnusamy VK, Jeyakumar RB, Sirohi R, Piechota G, Shobana S, et al. Microalgae cultivation strategies using cost-effective nutrient sources: Recent updates and progress towards biofuel production. *Bioresour Technol*. 2022;361:127691.
- Abdullah M, Ali Z, Yasin MT, Amanat K, Sarwar F, Khan J et al. Advancements in sustainable production of biofuel by Microalgae: Recent insights and future directions. *Environ Res*. 2024:119902.
- Mendoza Martin J. Raceway system requirements for low-cost energy-efficient algal biomass cultivation. University of Southampton; 2016.
- Penloglou G, Pavlou A, Kiparissides C. Recent advancements in photo-bioreactors for microalgae cultivation: a brief overview. *Processes*. 2024;12(6):1104.
- Razzak SA, Bahar K, Islam KO, Haniffa AK, Faruque MO, Hos-sain SZ, et al. Microalgae cultivation in photobioreactors: sustainable solutions for a greener future. *Green Chemical Engineering*. 2024;5(4):418–39.
- Mehmood MA, Amin M, Haq MAU, Shahid A, Malik S, Siddiqui AJ, et al. Assessment of molecular and metabolic traits of a newly isolated *Spirulina platensis* BERC15 in a low-cost cultivation alternative for its use as functional food. *Bioresource Technol Rep*. 2024;26:101816.
- Satiro J, Gomes A, Florencio L, Simões R, Albuquerque A. Effect of microalgae and bacteria inoculation on the startup of bioreactors for paper pulp wastewater and biofuel production. *J Environ Manage*. 2024;362:121305.
- Sátiro J, Fernandes JP, Mucha AP, Gomes AP, Calheiros CSC, Muñoz R, et al. Effect of slow feeding and hydraulic retention time on algal-bacterial aerobic granular sludge for low-nitrogen pulp mill wastewater treatment. *J Environ Chem Eng*. 2025. <https://doi.org/10.1016/j.jece.2025.118216>.
- Tang DYY, Khoo KS, Chew KW, Tao Y, Ho S-H, Show PL. Potential utilization of bioproducts from microalgae for the quality enhancement of natural products. *Bioresour Technol*. 2020;304:122997.
- Vuppalladadiyam AK, Prinsen P, Raheem A, Luque R, Zhao M. Microalgae cultivation and metabolites production: a comprehensive review. *Biofuels Bioprod Biorefin*. 2018;12(2):304–24.
- Singh G, Patidar S. Microalgae harvesting techniques: a review. *J Environ Manage*. 2018;217:499–508.
- Hernández A, González-Moya M, Márquez A, Acevedo LE. Review microalgae drying: a comprehensive exploration from conventional air drying to microwave drying methods. *Future Foods*. 2024. <https://doi.org/10.1016/j.fufo.2024.100420>.
- Corrêa PS, Morais WG Junior, Martins AA, Caetano NS, Mata TM. Microalgae biomolecules: extraction, separation and purification methods. *Processes*. 2020;9(1):10.
- Hasan M, Mofijur M, Uddin M, Kabir Z, Badruddin IA, Khan TY. Insights into anaerobic digestion of microalgal biomass for enhanced energy recovery. *Front Energy Res*. 2024;12:1355686.
- Zydney AL. Continuous downstream processing for high value biological products: a review. *Biotechnol Bioeng*. 2016;113(3):465–75.
- Deepa P, Sowndhararajan K, Kim S. A review of the harvesting techniques of microalgae. *Water*. 2023;15(17):3074.

38. Imamoglu E. Artificial intelligence and/or machine learning algorithms in microalgae bioprocesses. *Bioengineering Basel*. 2024;11(11):1143.
39. Zhou S, Jiang J, Hong X, fu P, Yan H. Vision meets algae: a novel way for microalgae recognition and health monitor. *Front Mar Sci*. 2023. <https://doi.org/10.3389/fmars.2023.1105545>.
40. Pattanaik A, Sukla L, Pradhan D, Shukla V. Artificial Intelligence and Virtual Environment for Microalgal Source for Production of Nutraceuticals. 2019;13.
41. Liyanaarachchi VC, Nishshanka GKSH, Sakarika M, Nimarshana PHV, Ariyadasa TU, Kornaros M. Artificial neural network (ANN) approach to optimize cultivation conditions of microalga *Chlorella vulgaris* in view of biodiesel production. *Biochem Eng J*. 2021;173:108072.
42. Trivedi J, Singh J, Atray N, Ray S, Agrawal D. Development of a non-linear growth model for predicting temporal evolution of *Scenedesmus obliquus* with varying irradiance. *Bioprocess Biosyst Eng*. 2019;42:2047–54.
43. Musa M, Ayoko GA, Ward A, Rösch C, Brown RJ, Rainey TJ. Factors affecting microalgae production for biofuels and the potentials of chemometric methods in assessing and optimizing productivity. *Cells*. 2019;8(8):851.
44. Zhang L, Liu J, Shen X, Li S, Li W, Xiao X. Response surfaces method and artificial intelligence approaches for modeling the effects of environmental factors on chlorophyll a in *Isochrysis galbana*. *Microorganisms*. 2023;11(8):1875.
45. Coşgun A, Günay ME, Yıldırım R. Machine learning for algal biofuels: a critical review and perspective for the future. *Green Chem*. 2023;25(9):3354–73.
46. Hossain SZ, Sultana N, Jassim MS, Coskuner G, Hazin LM, Razzak SA, et al. Soft-computing modeling and multiresponse optimization for nutrient removal process from municipal wastewater using microalgae. *J Water Process Eng*. 2022;45:102490.
47. Kushwaha OS, Uthayakumar H, Kumaresan K. Modeling of carbon dioxide fixation by microalgae using hybrid artificial intelligence (AI) and fuzzy logic (FL) methods and optimization by genetic algorithm (GA). *Environ Sci Pollut Res Int*. 2023;30(10):24927–48.
48. Ching PML, Mayol AP, San Juan JLG, Calapatia AM, So RH, Sy CL, et al. AI methods for modeling the vacuum drying characteristics of *Chlorococcum infusionum* for algal biofuel production. *Process Integr Optim Sustain*. 2021;5:247–56.
49. Ning H, Li R, Zhou T. Machine learning for microalgae detection and utilization. *Front Mar Sci*. 2022;9:947394.
50. Chong JWR, Tang DYY, Leong HY, Khoo KS, Show PL, Chew KW. Bridging artificial intelligence and fucoxanthin for the recovery and quantification from microalgae. *Bioengineered*. 2023;14(1):2244232.
51. Liyanaarachchi VC, Premaratne M, Nimarshana PHV, Ariyadasa TU, editors. Investigation of the effect of organic and inorganic carbon on biomass production and astaxanthin accumulation of the microalga *Haematococcus pluvialis* using artificial neural network. 2020 IEEE 17th India Council International Conference (INDICON); 2020: IEEE.
52. Sudfeld C, Kiyani A, Wefelmeier K, Wijffels R, Barbosa M, D'Adamo S. Expression of glycerol-3-phosphate acyltransferase increases neutral lipid accumulation in *Nannochloropsis oceanica*. 2022.
53. Yang W, Zhang H, Yang S, Xiao Y, Ye K, He R et al. Combined effects of microplastics and pharmaceutical and personal care products on algae: A critical review. *Environ Pollut*. 2024;124478.
54. Hasnain M, Abideen Z, Hashmi S, Naz S, Munir N. Assessing the potential of nutrient deficiency for enhancement of biodiesel production in algal resources. *Biofuels*. 2023;14(1):1–34.
55. Minzu V, Ifrim G, Arama I. Control of microalgae growth in artificially lighted photobioreactors using metaheuristic-based predictions. *Sensors (Basel)*. 2021;21:8065.
56. Long B, Fischer B, Zeng Y, Amerigian Z, Li Q, Bryant H, et al. Machine learning-informed and synthetic biology-enabled semi-continuous algal cultivation to unleash renewable fuel productivity. *Nat Commun*. 2022;13(1):541.
57. Ansari FA, Nasr M, Rawat I, Bux F. Artificial neural network and techno-economic estimation with algae-based tertiary wastewater treatment. *J Water Process Eng*. 2021;40:101761.
58. Wang CA, Onyeaka H, Miri T, Soltani F. *Chlorella vulgaris* as a food substitute: applications and benefits in the food industry. *J Food Sci*. 2024;89(12):8231–47.
59. Capelli B, Cysewski GR. Potential health benefits of *Spirulina* microalgae* A review of the existing literature. *Nutrafoods*. 2010;9(2):19–26.
60. Park E-K, Lee C-G. Astaxanthin production by *Haematococcus pluvialis* under various light intensities and wavelengths. *J Microbiol Biotechnol*. 2001;11(6):1024–30.
61. Camacho-Rodríguez J, Cerón-García M, Fernández-Sevilla J, Molina-Grima E. Genetic algorithm for the medium optimization of the microalga *Nannochloropsis gaditana* cultured to aquaculture. *Bioresour Technol*. 2015;177:102–9.
62. Sundui B, Ramirez Calderon OA, Abdeldayem OM, Lázaro-Gil J, Rene ER, Sambuu U. Applications of machine learning algorithms for biological wastewater treatment: updates and perspectives. *Clean Technol Environ Policy*. 2021;23:127–43.
63. Manochkumar J, Jonnalagadda A, Cherukuri AK, Vannier B, Janjaroen D, Ramamoorthy S. Enhancing cellular production of fucoxanthin through machine learning assisted predictive approach in *Isochrysis galbana*. *bioRxiv*. 2024:2024.04. 26.591373.
64. Izquierdo PF, Delgado LC, Benavides FO. An artificial neuronal network coupled with a genetic algorithm to optimise the production of unsaturated fatty acids in *Parachlorella kessleri*. *Artif Intell Agric*. 2024;13:32–44.
65. Olguín EJ, Sánchez-Galván G, Arias-Olguín II, Melo FJ, González-Portela RE, Cruz L, et al. Microalgae-based biorefineries: challenges and future trends to produce carbohydrate enriched biomass, high-added value products and bioactive compounds. *Biology Basel*. 2022;11(8):1146.
66. Selvam J, Mal J, Singh S, Yadav A, Giri BS, Pandey A, et al. Bioprospecting marine microalgae as sustainable bio-factories for value-added compounds. *Algal Res*. 2024;79:103444.
67. Ma X, Guan Y, Mao R, Zheng S, Wei Q. Modeling of lead removal by living *Scenedesmus obliquus* using backpropagation (BP) neural network algorithm. *Environ Technol Innov*. 2021;22:101410.
68. Ahmad Sobri MZ, Redhwan A, Ameen F, Lim JW, Liew CS, Mong GR, et al. A review unveiling various machine learning algorithms adopted for biohydrogen productions from microalgae. *Fermentation*. 2023;9(3):243.
69. Shah AM, Yang W, Mohamed H, Zhang Y, Song Y. Microbes: A hidden treasure of polyunsaturated fatty acids. *Front Nutr*. 2022;9:827837.
70. Song L, Qin JG, Su S, Xu J, Clarke S, Shan Y. Micronutrient requirements for growth and hydrocarbon production in the oil producing green alga *Botryococcus braunii* (Chlorophyta). *PLoS One*. 2012;7(7):e41459.
71. Yilmaz N, Yardimci CH, Elhag M, Dumitrache CA. Phytoplankton composition and water quality of Kamil Abduş Lagoon (Tuzla Lake), Istanbul-Turkey. *Water*. 2018;10(5):603.
72. Abu-Ghosh S, Dubinsky Z, Verdelho V, Iluz D. Unconventional high-value products from microalgae: a review. *Bioresour Technol*. 2021;329:124895.
73. Kraus J-I. Artificial intelligence applied to the production of high-added-value dinoflagellates toxins. *AI Soc*. 2020;35(4):851–5.

74. Shuai X, Zhan N, editors. AI-DRIVEN INNOVATIONS IN MICROALGAE BIOTECHNOLOGY: ENHANCING BIOFUEL PRODUCTION AND ENVIRONMENTAL HEALTH. The 3rd International scientific and practical conference Modern trends in the development of science and information technologies (September 17–20, 2024) Sofia, Bulgaria International Science Group 2024 263 p; 2024.
75. Akhter R, Masoodi F, Wani TA. Production and techno-characterization of *Chlorella vulgaris* microalgae as an alternative meat protein: Comparison with soya and pea proteins. *J Food Meas Charact.* 2026;1–17.
76. Patel P, Pang YLJ, Choi WJ, Wong A. Protein extraction and isolation from legumes and algae: An industry primer. *Food Bioprocess Technol.* 2025;18(10):8380–408.
77. Li A, Peng Y, Yao S, Zhang L, Meng F. Research on the application of machine learning in optimizing the heavy metal adsorption of algae-based adsorbents. *Algal Res.* 2026;104528.
78. Zhang Q, Lu Z, Liu Z, Li J, Chang M, Zuo M. Application of machine learning in food safety risk assessment. *Foods.* 2025;14(23):4005.
79. Hossain SMZ, Sultana N, Razzak SA, Hossain MM. Modeling and multi-objective optimization of microalgae biomass production and CO₂ biofixation using hybrid intelligence approaches. *Renew Sustain Energy Rev.* 2022;157:112016.
80. Mojiri A, Baharlouei M, Kazeroon RA, Farraji H, Lou Z. Removal of pharmaceutical micropollutants with integrated biochar and marine microalgae. *Microorganisms.* 2020;9(1):4.
81. Gorbachev V, Nikitina M, Velina D, Mutallibzoda S, Nosov V, Korneva G, et al. Artificial neural networks for predicting food antiradical potential. *Applied Sciences.* 2022;12(12):6290.
82. Garg S, Wang L, Schenk PM. Flotation separation of marine microalgae from aqueous medium. *Sep Purif Technol.* 2015;156:636–41.
83. Hawari AH, Alkhatib AM, Das P, Thaher M, Benamor A. Effect of the induced dielectrophoretic force on harvesting of marine microalgae (*Tetraselmis* sp.) in electrocoagulation. *J Environ Manage.* 2020;260:110106.
84. Liang C, Yang Y, Xia Y, Yuan W, Chen J, Zheng Z, et al. The optimization of *Chlorella vulgaris* flocculation harvesting by chitosan and calcium hydroxide. *Indian J Microbiol.* 2022;62(2):266–72.
85. Vázquez-Romero B, Perales JA, Pereira H, Barbosa M, Ruiz J. Techno-economic assessment of microalgae production, harvesting and drying for food, feed, cosmetics, and agriculture. *Sci Total Environ.* 2022;837:155742.
86. Hosseinizand H, Sokhansanj S, Lim CJ. Studying the drying mechanism of microalgae *Chlorella vulgaris* and the optimum drying temperature to preserve quality characteristics. *Drying Technol.* 2018;36(9):1049–60.
87. Silva NC, Duarte CR, Barrozo MA. Analysis of the use of a non-conventional rotary drum for dehydration of microalga *Spirulina platensis*. *Bioprocess Biosyst Eng.* 2020;43:1359–67.
88. Schmid B, Navalho S, Schulze PS, Van De Walle S, Van Royen G, Schüller LM, et al. Drying microalgae using an industrial solar dryer: a biomass quality assessment. *Foods.* 2022;11(13):1873.
89. Xiong Q, Pang Q, Pan X, Chika AO, Wang L, Shi J, et al. Facile sand enhanced electro-flocculation for cost-efficient harvesting of *Dunaliella salina*. *Bioresour Technol.* 2015;187:326–30.
90. Anyanwu RC, Rodriguez C, Durrant A, Olabi AG. Optimisation of tray drier microalgae dewatering techniques using response surface methodology. *Energies.* 2018;11(9):2327.
91. Eliseus A, Putra Z, Bilad M, Nordin N, Wirzal M, Jaafar J, et al. Energy minimization of a tilted panel filtration system for microalgae filtration: Performance modeling and optimization. *Algal Res.* 2018;34:104–15.
92. Zhang B, Peng C, Zhang S, Zhang M, Li D, Wang X, et al. Comprehensive analysis of the combined flocculation and filtration process for microalgae harvesting at various operating parameters. *Sci Total Environ.* 2023;857:159658.
93. Disconzi FP, Borghi FT. Modeling, simulation, and optimization of the centrifugal separation of a microalgae-culture medium mixture. *Biomass Bioenergy.* 2020;143:105871.
94. Machado CA, Esteves AF, Pires JC. Optimization of microalgal harvesting with inorganic and organic flocculants using factorial design of experiments. *Processes.* 2022;10(6):1124.
95. Rakesh S, Tharunkumar J, Sri B, Jothibasu K, Karthikeyan S. Sustainable cost-effective microalgae harvesting strategies for the production of biofuel and oleochemicals. *Highlights bioscience.* 2020;3(July):1–8.
96. Teng SY, Yew GY, Sukačová K, Show PL, Máša V, Chang J-S. Microalgae with artificial intelligence: a digitalized perspective on genetics, systems and products. *Biotechnol Adv.* 2020;44:107631.
97. Otálora P, Guzmán JL, Acien FG, Berenguel M, Reul A. Microalgae classification based on machine learning techniques. *Algal Res.* 2021;55:102256.
98. Islam Z, Khatoon H, Minhaz TM, Rahman MR, Hasan S, Mahmud Y, et al. Data on growth, productivity, pigments and proximate composition of indigenous marine microalgae isolated from Cox's Bazar Coast. *Data Brief.* 2021;35:106860.
99. Rubio C, Dominik-Jakubiec M, Paz S, Gutiérrez AJ, Gonzalez-Weller D, Hardisson A. Dietary exposure to trace elements (B, Ba, Li, Ni, Sr, and V) and toxic metals (Al, Cd, and Pb) from the consumption of commercial preparations of *Spirulina platensis*. *Environ Sci Pollut Res.* 2021;28(17):22146–55.
100. Saha R, Chauhan A, Rastogi Verma S. Machine learning: an advancement in biochemical engineering. *Biotechnol Lett.* 2024;46(4):497–519.

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